

Figure 1 is a scatter plot showing the temperature dependence of the critical current density  $J_c(T)$  for Bi2212. The x-axis represents Temperature [K] from 4 to 16, and the y-axis represents  $J_c(T)$  [A/cm²] from 0.0 to 20.0. Data points are shown for currents of 0.5, 0.8, 1.0, 1.5, 2.0, 2.5, 3.0, and 4.0 mA. A dashed line represents the fit  $J_c(T) = 0.105T^2 - 0.5315T + 0.8116$ . Open circles represent reference data from [10].

| Temperature [K] | $J_c(T)$ [A/cm²] (0.5 mA) | $J_c(T)$ [A/cm²] (0.8 mA) | $J_c(T)$ [A/cm²] (1.0 mA) | $J_c(T)$ [A/cm²] (1.5 mA) | $J_c(T)$ [A/cm²] (2.0 mA) | $J_c(T)$ [A/cm²] (2.5 mA) | $J_c(T)$ [A/cm²] (3.0 mA) | $J_c(T)$ [A/cm²] (4.0 mA) |
|-----------------|---------------------------|---------------------------|---------------------------|---------------------------|---------------------------|---------------------------|---------------------------|---------------------------|
| 4.5             | 0.5                       |                           |                           |                           |                           |                           |                           |                           |
| 5.2             |                           | 0.8                       |                           |                           |                           |                           |                           |                           |
| 5.8             |                           |                           | 1.5                       |                           |                           |                           |                           |                           |
| 7.5             |                           |                           |                           | 2.8                       |                           |                           |                           |                           |
| 9.2             |                           |                           |                           |                           | 4.8                       |                           |                           |                           |
| 10.8            |                           |                           |                           |                           |                           | 7.5                       |                           |                           |
| 12.5            |                           |                           |                           |                           |                           |                           | 10.5                      |                           |
| 15.5            |                           |                           |                           |                           |                           |                           |                           | 18.0                      |

Fit:  $y = 0.002984x^2 + 0.1156x - 0.4448$

| Temperature [K] | Current [mA] | y-value |
|-----------------|--------------|---------|
| 4.5             | 0.5          | 0.1     |
| 5.2             | 0.8          | 0.25    |
| 5.8             | 1.0          | 0.35    |
| 7.5             | 1.5          | 0.6     |
| 9.2             | 2.0          | 0.9     |
| 10.8            | 2.5          | 1.15    |
| 12.5            | 3.0          | 1.45    |
| 15.5            | 4.0          | 2.1     |